Technical Datasheet

Autonomous Greenhouse Harvesting Robot - v1.0 - 2026

Specifications below describe the E-Farm platform. Values marked (target) or (illustrative) are design goals for the first production batch and may change; verified figures are from our project test set and selected components.

Perception & Vision

Parameter	Specification
Vision model	YOLOv8 object detection (trained via Roboflow)
Training data	10,000+ annotated images across 5 crops
Ripeness accuracy	98% on project test set (verified, internal)
Detected crops	5 greenhouse crops (e.g. tomato, pepper, cucumber)
Depth sensing	RGB-D camera + LiDAR fusion; sub-cm localisation (target)

Manipulation

Parameter	Specification
Manipulator	6-axis arm, PAROL6-based (open-source, GPLv3)
Repeatability	+/- 0.1 mm class (illustrative, design class)
Reach	~440 mm (target)
Payload	~1 kg (target)
End-effector	Adaptive gripper, force-limited soft grasp

Mobility & Chassis

Parameter	Specification
Chassis	NASA JPL Open Source Rover (OSR)-derived (illustrative)
Drive	Multi-wheel rocker-style platform for uneven greenhouse floors
Navigation	Vision + LiDAR aided, structured-row greenhouse operation

Power & Electronics

Parameter	Specification
Battery	LiFePO4 pack (safe, long-cycle chemistry)
BMS	ESP32-based battery management system
Runtime	~8 h per charge (target)

Software & Connectivity

Parameter	Specification
OS / middleware	ROS2 Humble
Telemetry / fleet	MQTT messaging; multi-robot fleet dashboard
Architecture	Perception -> harvest FSM -> arm -> telemetry

Connectivity	Wi-Fi / Ethernet; MQTT broker uplink
Source	Open-source monorepo: github.com/meviza/E-Farm-Autonomous-Harvesting-Robot

Physical (illustrative / target)

Parameter	Specification
Footprint	~700 x 600 mm (illustrative)
Height	~1,300 mm with arm stowed (illustrative)
Mass	~45 kg (illustrative)
Operating env.	Protected greenhouse, 0-40 deg C, high humidity

Note on figures

Items labelled (target) or (illustrative) are pre-production design goals. The vision accuracy figure is measured on our internal project test set. PAROL6 is licensed GPLv3; E-Farm's own designs are published open-source in the project monorepo.